SAINT MICHAEL COLLEGE OF CARAGA

FOOD SAFETY PRACTICES OF STREET FOOD VENDORS IN BUENAVISTA AND NASIPIT

A Thesis Proposal Presented To
The Faculty of
College of Tourism and Hospitality Management

In Partial Requirements for the Degree
BACHELOR OF SCIENCE IN HOSPITALITY MANAGEMENT

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CHAPTER 1

THE PROBLEM AND ITS SCOPE

1.1 Background of the Study

Street food vending will be a vibrant aspect of the global informal food economy. It will offer cheap and convenient meals for millions and will provide a means of earning a living for low-income individuals. However, it will also offer significant food safety problems because of poor infrastructure, poor hygiene training, and poor regulation (WHO, 2021). According to the World Health Organization, an estimated one in every ten individuals will become ill every year from foodborne contamination, resulting in 420,000 deaths worldwide. Foodborne illness will continue to place significant strains on public health systems, particularly in developing nations where street foods will provide students and workers with meals (Rane, 2011; Mensah & Owusu, 2021). Street food culture will remain rooted in the daily life of Filipinos—from urban areas to small towns like Buenavista and Nasipit in the province of Agusan del Norte. The municipalities under study will experience a significant increase in informal street food vending, particularly in school areas, transport terminals, and public parks. Despite its economic and cultural significance, researchers know little will be known about whether these vendors will observe proper food safety practices. Hence, this study will be timely and relevant, as it will aim to examine food safety management practices and identify the underlying factors contributing to non-compliance with hygienic standards.

The researchers will base this study on the Hygiene and Sanitation Theory. Wasike (2010) emphasizes that researchers must critically analyze hygiene and sanitation practices in the hospitality and tourism industry to identify gaps between theory and practice. This theory will allow researchers to consider the factors involved in street food vending, which will require high standards and suitable locations, and will

serve as a valuable framework for promoting internal hygiene and sanitation. According to Osei-Mensah et al. (2022), they were moderately knowledgeable but had infrastructure constraints in Ghana. Meanwhile, Nigerian and Ghanaian studies will point out unhygienic conditions, emphasizing education and income (Chukuezi, 2010; Akabanda et al., 2017). In Bangladesh, studies will reveal a positive relationship between awareness, experience, and education. However, practice will likely remain patchy (Rahman et al., 2023).In India and Nepal, researchers will observe poor food handling and minimal use of protective equipment (Das et al., 2019; Thapa et al., 2020). Malaysian government training will enhance vendor compliance (Mohd et al., 2018), while Tanzanian and Kenyan studies will indicate unhygienic environments and sporadic inspections (Mwangi et al., 2021; Nyonyo et al., 2020). The combined studies will demonstrate that even with awareness, adherence to safety practices will not be constant because of infrastructural and socioeconomic challenges. Age and experience will also affect hygiene practices (Arku et al., 2020; Mensah & Owusu, 2021). Research shows that street foods have been known to contribute to the spread of diseases, such as those caused by the contamination of foods and the poor practices of personal and kitchen hygiene. Bacterial contaminants such as E. coli, Salmonella, Shigella, Staphylococcus aureus, Listeria, and Bacillus cereus have been shown to cause diseases such as gastroenteritis (Andrade et al., 2023; Fröder et al., 2021; Bonomo, 2023).

Locally, a Belgica (2024) study will find that Estancia, Iloilo, vendors will have good theoretical knowledge of sanitation but will lack practical implementation under constrained resources. Raneses et al. (2018) will note that market vendors in Metro Manila will possess good knowledge but will lack proper food handling and waste disposal. Guinto et al. (2022) in Biliran Province also found that vendors demonstrated positive attitudes but were unaware of contamination hazards. However, in Zamboanga

del Norte, it emphasized that the gap between awareness and consistent hygienic practices remains a serious challenge, street food vendors' sanitation practices were described as being "often practiced," that is, as showing moderate adherence to the correct standards of hygiene. The study enumerated twenty detailed sanitation practices among Dipolog and Dapitan city vendors. These included either food preparation at home or on-site, protection of vending stalls from wind, sun, and dust, and restriction of animals or rodents such as flies around vending places. Vendors also received clean waste places of disposal, repaired damaged utensils promptly, and segregated chemicals such as cleaning agents from foods. Some of the regular practices were cleaning raw foods prior to cooking, utensils cleaned between use with sanctioned washing agents, and they were stored in covered containers. In addition, vendors also removed cracks, rust, and from cupboards, ceilings, walls, and floors, and maintained benching, sinks, shelves, and tables in good condition. Some of these acts, however, were also found to be unhygienic in nature, such as having open food lying around in stalls, chopping raw and cooked food with the same knives and chopping boards, and using the same washing basins by consumers and food handlers. Optimal as they were, these practices were still at risk of contamination, and hence more stringent sanitary oversight and regular food safety training for vendors were implied (Solon, 2022). Food handlers and waste managers frequently apply conventional means and do not have formal training on food handling and waste management. Likewise, Limon et al. (2021) in Cebu City revealed that school and terminal vendors seldom utilized gloves or hairnets even if they knew their necessity. Among street food vendors in Zamboanga del Norte, according to the research of Solon (2022), the vendors normally observed good personal hygiene. They frequently washed their hands when preparing food and upon going to the toilet, used clean attire and aprons, and had short and clean nails. Some

wore gloves, hairnets, and masks when touching food. Nevertheless, some vendors continued to handle food with bare hands, labor while sick, or work while handling money while serving food, presenting hygiene issues. On a general note, Solon underlined that although awareness of hygiene was apparent, its constant practice and observation were needed in order to guarantee food safety. These studies show that although government campaigns and school programs have enhanced food safety consciousness, sustainable compliance remains missing, especially in semi-urban and rural zones such as Buenavista and Nasipit.

This study is essential to localize information regarding the food safety behavior of street food vendors in Buenavista and Nasipit, with minimal existing research. Earlier research in other parts of the Philippines, such as Estancia, Iloilo (Belgica, 2024), Zamboanga del Norte (Solon, 2022), Catarman, Northern Samar (Madario et al., 2023), Alfonso Lista, Ifugao (Ifugao State University, 2023), and Dipolog City (Benitez & Olmogues, 2023), all find that inadequate infrastructure, insufficient training, and lax enforcement of good food safety practices restrict vendors despite their sufficient knowledge. International research corroborates these results, including those in Ghana (Nortey et al., 2024), India (Mukherjee et al., 2018), Malaysia (Abdul Aziz et al., 2023), and Vietnam (Nguyen et al., 2022), highlighting how age, education, and perceived behavioral control (Ajzen, 1991) influence vendor conduct. The study provides evidence for government health experts to create placements that address training, sanitation, and hygiene gaps. The LGUs can use the data to design regulations and support mechanisms appropriate to local vendors. The vendors will benefit through improvement in practice, consumer trust, and revenues. Implementing proper food safety practices will better protect consumers from foodborne diseases. Lastly, the study

adds to academic scholarship by filling a gap in Mindanao's semi-urban food safety research.

This study has essential suggestions for stakeholders, especially public health, local government, and the informal food sector. It provides evidence-based findings on street food vendors' food safety practices in Buenavista and Nasipit, which can guide public health practitioners and LGUs in crafting targeted interventions, policies, and training programs.

1.2 Theoretical Framework

This research is based on Wasike's (2010) Theory of Hygiene and Sanitation, which holds that a clean and safe environment is crucial in preventing microbial presence and ensuring food safety. The theory avers that hygiene and sanitation are two concepts that go hand in hand and are critical in foodborne disease prevention and that the closure of the gap between theoretical knowledge and practical application is key to successful food handling. In the context of street vending where vendors usually work within open spaces with limited availability of clean water, waste management facilities, and sanitation facilities this theory is especially applicable. It offers a framework for examining how cleanliness of the environment, individual hygiene, and institutional support all impact food safety behaviors. This was in agreement with Solon (2022), who discovered that hygiene and sanitation practices are still the primary determinants of food safety for street food vendors in the twin cities of Zamboanga del Norte, highlighting that malnutrition and foodborne illnesses are on the increase despite the presence of policies and laws. It was reported by Rahman et al. (2021) that vendors in Bangladesh had limited awareness of hygiene standards with resulting unsafe handling food behaviors, while Tawodzera, Chagonda (2022) described poor infrastructure and

limited enforcement of food safety laws as key issues among vendors in Harare. Odonkor, Ampofo (2020) further noted that while vendors know the principles of food safety, external constraints like resource limitations and supervision impede the consistent enactment of hygiene measures. Supporting these results, Nguyen et al. (2023) found that specific food safety training and frequent inspections greatly enhanced street vendors' hygiene behavior in Vietnam, illustrating that theoretical concepts can result in tangible gains with the support of education and enforcement. These studies point out that hygiene and sanitation theories give a robust conceptual framework but the practical issues especially in the informal food industries mostly prevent their application. The current research employs the Hygiene and Sanitation Theory as a conceptual framework to assess the perceived sanitary and hygienic practices of street food vendors in Nasipit and Buenavista, with the objectives of exploring how theoretical ideals are practiced daily, determining areas for gap between knowledge and behavior, and identifying factors affecting compliance with food safety standards. This framework is the foundation for suggesting measures to enhance regulation, improve vendor training, and develop a culture of sanitation and hygiene that supports public health and food safety in street food vending operations.

1.3 Review of Related Literature

This chapter will analyze past studies and research on food safety practices of street food vendors in Buenavista and Nasipit. It will discuss the different variables that affect food safety. The chapter will consider vendor information, food handling practices, surroundings, regulatory adherence, and how these factors relate. From this understanding of relations, the chapter will lay the ground for analyzing how these components impact overall street food safety in these regions. The review will discuss

the challenges and opportunities that emerge locally, seek to enhance the comprehension of food safety among vendors, and propose possible improvements.

Related Literature

Street vending of food is an important source of income and readily available food for municipalities such as Buenavista and Nasipit, Agusan del Norte, particularly in urban settings. Food safety, however, continues to be a topmost issue, largely blamed on the lack of regulation of the vending business coupled with loopholes in laws and irregular compliance with stipulated safety standards. It is not a peculiar problem in the Philippine context, as proven by research done in other varied global settings. Example includes that of Solon (2022), who studied food safety behaviors among street food vendors of Zamboanga del Norte twin cities. Adem (2023) adopted a cross-sectional design to investigate food hygiene practice and determinants among street food vendors in northeastern Ethiopia. Similarly, Negassa (2023) investigated food hygiene practice and determinants among urban street food vendors of the Gedeo Zone, southern Ethiopia. Chekol (2021) studied food safety practice and determinants associated with street food vendors in West Gojjam Zone urban councils, northwest Ethiopia. Hassan (2020) compared food hygienic practice and food safety control among street food vendors in the Zanzibar Urban District. Chopra (2021) evaluated the knowledge, educational level, and food handling practices of street food vendors critically. Cataluna (2024) conducted a systematic review of food hygiene and sanitation knowledge and practices of street food vendors. Wallace (2022) mapped vendor knowledge, attitudes, and practice towards food safety in low- and middle-income countries. Tamiru (2022) studied food safety behavior and determinants of food handlers working in food facilities of Mettu and Bedelle towns in southwest Ethiopia. Desye (2023) had a systematic review and meta-analysis of food safety attitudes,

knowledge, and practices of street food vendors and related factors in low- and middle-income countries. Mwove (2020) conducted an investigation of food safety knowledge and food vendors' practices at the chosen points of Kiambu County, Kenya. Nortey (2024) conducted a cross-sectional study to identify knowledge, attitude, and food safety practice among food vendors within a city of Ghana. Islam et al. (2024) carried out poor food safety awareness and hygiene practice among street food vendors in Dhaka, Bangladesh. Ghosh (2023) wrote on hygienic practice and street food safety along medical college areas in Kolkata, India. Gonzales (2023) had knowledge, attitudes, and practice of food handling among the Biliran Province street food vendors. Addo-tham (2020) examined the food safety knowledge and food-handling practices of Ejisu-Juaben Municipality's street food vendors in Ghana. Madilo (2022) examined food handling knowledge and practice among school food handlers. Abid (2022) evaluated Chattogram street food vendors' knowledge, attitudes, and practice of food safety in Bangladesh. Moges et al. (2023) evaluated sanitary condition and awareness on food safety among Ethiopian street vendors. Azanaw (2022) conducted a research on food safety practice and awareness among northern Ethiopian street vendors. Each of these studies, respectively, suggests that vendors do not typically possess proper knowledge of food safety practices and how to use them practically. These vulnerabilities are frequently associated with infrastructural shortcomings, such as inaccessibility of clean water and toilets, and poor training facilities. Yi and Choi (2025) studied consumers' food safety knowledge, attitudes, and practices with different ages in Seoul. Almutlaq et al. (2025) conducted a food knowledge and preferred sources of awareness survey among Saudi society. Milovanova et al. (2024) discussed the challenges to food safety understanding through a social and gender perspective in low- and middle-income countries. Garsow et al. (2022) investigated the interaction between gender and dairy

value chain food safety risks in Ethiopia. Zanin et al. (2021) investigated the effect of educational interventions on the transition of food safety culture in an army food service environment. Chen et al. (2025) examined sound food-handling practice factors among male food handlers working in catering establishments. Borbon and Tolentino (2020) examined food safety practices of food establishments as a foundation for improving policy. Agrawal et al. (2025) analyzed street food safety, emphasizing the hygiene behavior and risk factors among street vendors. Huynh-Van et al. (2022) investigated determinants of food safety compliance among street food vendors in Can Tho city, Vietnam, and their role in planning intervention activity development and execution. Dollentas and Habla (2021) discussed food hygiene and food safety practices among food businesses. Limon (2021) investigated food safety practices of household food handlers who are operating online food ventures during the COVID-19 pandemic in the Philippines. Origenes et al. (2022) conducted a study on microbiological load of street food and food hygiene and safety practices compliance of vendors in schools of Cebu City. Caye (2024) investigated food safety awareness and practice of street food vendors in some barangays in Marawi City. Jose and Villanueva (2023) assessed Calapan City's street food vendors' food safety dedication and hygiene practice. Rifat et al. (2022) conducted a systematic review of food handlers' food safety knowledge and practice in Bangladesh. The Knowledge, Attitude, and Practice (KAP) model supposes that even with adequate knowledge and positive attitude of vendors towards food safety, lack of finances and regulatory loopholes may prevent the effective practice of safe practice on a consistent basis. The Department of Science and Technology (DOST) has recently initiated food safety awareness campaigns in Buenavista. However, there is an apparent research gap regarding street vendors' food handling practices in Nasipit.

This lack of empirical evidence highlights the urgency for more research in order to understand street food vendors' operations and issues regarding safe handling of food. Filling this gap, the current study is set to compare food safety practices in Buenavista and Nasipit with lessons learned on the effectiveness of public health interventions and policy implications for local government to further reinforce food safety in the informal food market. By using systematic examination of these practices, the current study seeks to make evidence-based contributions towards improved levels of food safety and protection of community health. Drawing on the foundational knowledge created by the previous studies, it is important to have a more detailed exploration of particular themes and research approaches. There have been various studies that have identified the particular knowledge gaps responsible for unsafe food handling practices. For instance, there has been consistent evidence in studies showing a lack of knowledge on how to wash hands properly, how to control temperature, and how to prevent cross-contamination (Addo-tham, 2020; Islam et al., 2024; Moges et al., 2023). All these areas of lack of knowledge tend to be compounded by insufficient training and educational material, especially in low-resource environments. Aside from the issues of knowledge, attitudinal elements also contribute significantly to how food safety behaviors are developed. Research has indicated that vendors' risk perceptions, their attitudes toward the significance of food safety, and their risk attitudes in compliance with regulations can have considerable impact on vendors' practices (Wallace, 2022). For example, retailers with low priority perception of food safety or those perceiving rules and regulations to be too stringent might be less likely to adopt recommended guidelines. Attitudinal barriers knowledge is required for informing effective interventions that lead to behavioral change. The use of the KAP model has helped elucidate the dynamic interrelationship among knowledge, attitude, and practices

under street food vending. Although knowledge and attitude are necessary precursors for safe food handling, they are not always translated into habitual practice. Economic constraints, such as the cost of introducing food safety interventions, and regulatory loopholes, such as the absence of enforcement, are potential adoption-inhibiting barriers to safe practices (Desye, 2023). Addressing these system barriers is vital in ensuring an enabling environment for safe food vending.

In addition, other research has identified the role of socio-cultural factors in food safety intervention design and delivery. Cultural beliefs, handling practices, and gender roles may all affect food safety conduct (Milovanova et al., 2024; Garsow et al., 2022). Interventions that are culture-specific and address the unique needs and conditions of specific populations will be more successful. The geographical specificity of current studies also points to key areas in which our knowledge of street food vending food safety remains insufficient. While research has been carried out in many parts of the world, including Africa, Asia, and Latin America, there is a comparative lack of research in some areas, like Southeast Asia and the Pacific Islands. Furthermore, very few studies have explored in particular the challenges and opportunities that are peculiar to street food vending in island countries, like the Philippines. The current research expands on this existing literature by considering the particular case of Buenavista and Nasipit, Agusan del Norte. These municipalities are a microcosm of the greater issues and opportunities involved with street food vending in the Philippines. Through cross-comparison of the food safety practice within these two municipalities, the study aims to learn about the effectiveness of public health interventions and offer policy recommendations for local government. The research will make use of the mixed-methods design, combining surveys with quantitative data and interviews with qualitative data, in order to collect data on knowledge, attitudes, and practices among

vendors towards food safety. Surveys will measure vendors' knowledge of basic food safety principles, their attitude towards food safety regulation, and their reported practices. Interviews will give a better picture of the challenges and opportunities vendors encounter in practicing safe food handling habits. Study results will be utilized to create evidence-based recommendations on enhancing food safety in street food vending in Buenavista and Nasipit. These suggestions can involve specific training schemes, regulatory reforms, and community interventions. The research will also add to the overall pool of information on food safety in street food vending and lessons that can be used for other municipalities across the Philippines and other places. This research will have numerous contributions to food safety and public health. First, it will conduct a thorough evaluation of street food vending food safety practices in Buenavista and Nasipit, Agusan del Norte. Second, it will clarify the critical factors that determine vendors' food safety behaviors such as knowledge, attitudes, economic limitations, and regulatory loopholes. Third, it will formulate evidence-based interventions to enhance food safety among street food vending in these municipalities. Lastly, it will be part of the overall body of knowledge regarding street food vending's food safety, something that can be shared with other municipalities within the Philippines and overseas. By highlighting the gap in the research on street vendors' food handling practices in Nasipit and how they compare with the same practices in Buenavista, this study will provide useful insights into the efficacy of public health interventions and policy implications for local governments to improve food safety in the informal food market, ultimately protecting community health.

Sanitation Practices

Sanitation practices are being considered critical global public health determinants, whereby their absence creates extreme health dangers. Werkneh (2023)

provided an investigation on food handling habits and knowledge level among street food vendors in Mekelle City, Ethiopia, hence highlighting the necessity for increased hygiene norms in food production paradigms. Girma (2024) also interrogated the food safety habits and attitudinal dispositions among street vendors in Ghana, highlighting the central position of vendor behavior toward guaranteeing food safety standards. Wu (2024) performed a microbiological quality evaluation of street-vended foods in China, thereby unravelling potential contamination scenarios inherent in street food delivery. Nortey (2024) carried out a comparative evaluation of the knowledge, attitudes, and food safety practices among street food vendors within Ghana's city boundaries, realizing lacunae in their comprehension and compliance with laid down safety protocols. Kaddouri (2025) contrasted food safety behaviors with the microbiological quality characteristics of street foods in Marrakech, Morocco, highlighting disparity in food safety standards across different working environments. Sharma (2025) analyzed the hygiene practices of urban street food vendors with a focus on the need for intervention intended to promote better hygiene practices. Koumassa (2025) laid down the determinants that determine the quality and safety standards of street-vended food in developing nations, with insights on the intricacies involved in food safety in these nations. Carpio (2025) delved into the practices, knowledge, and attitudes of street food vendors in the Philippines towards food safety, with due emphasis on the use of culturally sensitive strategies towards food safety education initiatives. Klutse (2025) analyzed the standards of hygiene and the standards of food safety knowledge of street food vendors in Ghana and thus reiterated the importance of regular training and vigilant monitoring systems. Girma (2025) also critically examined food safety and microbial contamination phenomena in Ethiopian street food vendors, emphasizing the significance of adapting to food safety imperatives in the Ethiopian environment. A 2025

report offered an overall view of the hygiene and safety compliance shown in street food vending and emphasized the need for regular enforcement of regulatory standards. Kamboj (2020) provided a food safety and hygiene review, thus giving a comprehensive overview of the cardinal principles and practices that guide these aspects. Wiatrowski (2021) examined the food vending street vendors' hygiene practices among Polish food vendors working in approved food stalls and food trucks, adding to the existing body of knowledge in food safety practices within the European context. Begum (2024) surveyed the microbial levels of contamination of street food as well as the sanitation conditions of vendors in Bangladesh, thus highlighting the issues involved in maintaining food safety standards in populous areas. Carpio (2024) also carried out microbiological analysis of street foods and water samples in Ethiopia, providing additional evidence of possible vectors of contamination. Lista (2025) assessed the knowledge, attitudes, and practices related to food safety of street food vendors at Alfonso Lista, Ifugao, Philippines, thus enriching local knowledge on food safety demands. Loganathan (2025) evaluated street food vending stalls in urban and semi-urban Puducherry that adhered to food safety and hygiene practices, indicating the necessity of custom-designed interventions for diverse urban typologies. Aduol (2020) evaluated food street-vending practices under the parameters of hygiene and microbial contamination in Kenyatta University precincts with focus on the specific challenges of academic environments. Yi and Choi (2025) contrasted age differences in consumer knowledge, attitude, and practice towards food safety in Seoul, illustrating how individuals of different ages may possess distinct levels of sensitivity and compliance with food safety laws. Almutlaq et al. (2025) evaluated awareness of foods and preferred modalities for delivery of awareness among Saudi individuals, thus identifying effective communication media for the delivery of food safety awareness.

Milovanova et al. (2024) talked about food safety issues from the social and gendered perspective in low- and middle-income countries, and it was noted that there is a need to consider gender dynamics when planning food safety interventions. Garsow et al. (2022) undertook studies on the intersection of gender and food safety risk within the Ethiopian dairy value chain with a view to examining gendered risk exposures in farming settings. Islam et al. (2024) identified poor food safety knowledge and hygiene practices among street food vendors in Dhaka, Bangladesh, hence the need for specialized training and education interventions. Zanin et al. (2021) analyzed the impact of education interventions on food safety culture building within the context of an army catering facility and contended that educational interventions hold the potential to greatly enhance food safety processes in the context of institutions. Chen et al. (2025) discussed factors associated with exemplary food-handling habits in male food handlers working in catering outlets, thus providing evidence on the determinants of safe food handling habits. Borbon & Tolentino (2020) weighed on the food safety culture of food establishments as the basis for policy improvement, highlighting the regulatory function of promoting and enforcing food safety. Moges, Rødland, and Argaw (2023) examined the sanitary standard of street food vendors in sample Ethiopian towns, upholding the need for enhanced sanitation facilities and hygiene measures. Desye et al. (2023) carried out a systematic review and meta-analysis of street food vendors' knowledge, attitude, and practices regarding food safety in low- and middle-income countries, an overview of the challenges and opportunities for improving food safety in these countries. Agrawal et al. (2025) created a study explaining hygiene practice and risk factors among the vendors, hence giving insights into the particular risk factors leading to foodborne illness. Huynh-Van et al. (2022) examined factors that predict food safety compliance among street food vendors in Vietnam's Can Tho city, providing important

information for the plan and execution of successful interventions. Dollentas & Habla (2021) concentrated on food hygiene and food safety practices in food premises, with special emphasis placed on maintaining high standards of food handling and preparation procedures. Limon (2021) examined the food safety conduct of food handlers who partake in online food operations amid the COVID-19 pandemic, thus putting into focus the distinctive challenges and hazards of online food delivery modalities. Origenes, Espinosa-Gelisanga, and Alturas (2022) investigated the microbiological load and food hygiene conduct of street food vendors in schools of Cebu City, highlighting the need for tight regulation and monitoring in educational settings. Caye (2024) determined food safety awareness and practices of street food vendors in Marawi City, thus presenting localized evidence for the development of targeted interventions. Solon (2022) examined food safety practices of street food vendors in Zamboanga del Norte, adding to regional variations in food safety practices. Jose & Villanueva (2023) evaluated the hygiene practices and food safety commitment of Calapan City street food vendors in terms of relevance to continuous monitoring and training programs. Rifat et al. (2022) carried out a systematic review of food safety awareness and practice of food handlers in Bangladesh and therefore provided an overall scenario of food safety trends in the Bangladeshi setting. These collective scholarly works highlight the multidimensional nature of food safety concerns involved and the resultant need for integrated, context-based interventions to strengthen sanitation measures and mitigate foodborne disease risks.

Personal Hygiene

Personal hygiene is among the pillars of public health that play an important role in infectious disease risk and well-being of the global population. Numerous research studies worldwide have time and again identified food safety KAP's pivotal

relevance towards preventing foodborne illness. These studies, done in various environments like street food vending, food shops, and homes, together present an evidence-based view of food safety issues and possible intervention measures.Studies among street food vendors, who sell much of the food for urban dwellers, have shown bad hygiene and safety behavior. Solon (2022), in "Food Safety Practices Among Street Food Vendors in the Twin Cities of Zamboanga del Norte," established where practices were below standard to get it safe in the Philippines. The same, Adem (2023) carried out a cross-sectional study in Northeastern Ethiopia to determine factors on poor food hygiene among vendors. Negassa (2023) studied food hygiene practice and determinants among urban street food vendors within the Gedeo Zone, Southern Ethiopia. This research confirms intervention is necessary in special manners. Chekol (2021) contrasted street food vendors' food safety practices between city councils within the West Gojjam Zone, Northwest Ethiopia, and established differences to be considered. Hassan (2020) studied safe and clean behavior among street food vendors in the Zanzibar Urban District with an East African regional flavor. Islam et al. (2024), "Inadequate Food Safety Knowledge and Hygiene Practices Among Street Food Vendors in Dhaka, Bangladesh," found inadequate food safety knowledge and hygiene practice with the significance of improving them in densely populated urban areas. Ghosh (2023) recorded food hygiene practices and street food food safety near the medical colleges of Kolkata, India, yielding data on a delimited high-traffic area. Gonzales (2023) assessed food handling knowledge, practice, and attitude of street food vendors in Biliran Province in the Philippines, adding to local practices knowledge. Addo-Tham (2020) quantified food-handling ability and food safety knowledge for Ghana's Ejisu-Juaben Municipality street food vendors and established needs for skill development.Desye et al. (2023), "Food safety knowledge, attitude, and practice of street food vendors and

associated factors in low- and middle-income countries: A systematic review and meta-analysis," provided a comprehensive discussion of the problems and available solutions. Mwove (2020) carried out a test of food safety knowledge and practice of street food vendors in Kiambu County, Kenya, contributing to information from the African continent. Nortey (2024) recorded the food safety behavior, attitude, and knowledge of street food vendors in a Ghanaian city, providing up-to-date facts regarding the practices. Research on food facilities comes after food safety research on street food vendors, indicating food safety practices in formal facilities. Borbon & Tolentino (2020), in "Food Safety Practices of Food Establishments: Basis for Policy Enhancement," investigated food safety practices of food establishments in the Philippines as a basis for policy enhancement. Dollentas & Habla (2021), in "Food Hygiene and Safety Practices of Food Establishments," further investigated food hygiene and safety practices of food establishments as a contribution to the body of knowledge on practice in the food service industry. Chen et al. (2025), "Factors related to good food-handling practices: a cross-sectional study based on male food handlers in catering establishments," identified factors correlating with good food-handling practice among male food handlers within catering outlets, giving gender-specific practice findings.Household-level studies emphasize the significance of food safety practice and awareness in preventing foodborne disease at the household level. Rabeya et al. (2022) conducted a cross-sectional community-based survey with the rural communities of Bangladesh and assessed the KAP of rural communities with respect to food safety practices. Income and educational status were determined as determinants of food safety activity with respect to socioeconomic determinants. There are numerous other studies targeted towards various populations and settings that offer deeper insights into food safety activities. Yi and Choi (2025), "Age differences in food safety knowledge,

attitudes, and practices among consumers in Seoul," examined age differences in food safety knowledge, attitudes, and practices among Seoul consumers and their meaning for age-targeted intervention. Almutlaq et al. (2025), "Assessment of Food-Related Knowledge and Preferred Awareness Methods by the Saudi Community: A Cross-Sectional Study," assessed food-related knowledge and most preferable awareness methods in the Saudi community and emphasized the importance of culture-based communication strategies. Garsow et al. (2022), in "Exploring the relationship between gender and food safety risks in the dairy value chain in Ethiopia," analyzed gender-food safety risk nexus in the dairy value chain in Ethiopia and developed gender-based vulnerabilities.A systematic overview of knowledge, educational level, and food handling behavior among street food vendors was given by Chopra (2021). Cataluna (2024) conducted a systematic review of street food vendors' knowledge and practice of food sanitation and hygiene and areas for enhancement. Wallace (2022) conducted a scoping review of food safety knowledge, attitude, and practice of vendors in low- and middle-income countries and gave a worldwide overview. Zanin et al. (2021), in "Influence of educational actions on transitioning of food safety (FS)-culture: a longitudinal study using action research in an army food service setting," illustrated the impact of educational actions on transitioning food safety culture within an army food service environment. Agrawal et al. (2025), entitled "Street Food Safety: A Study on Hygiene Practice and Risk Factors among Street Food Vendors," studied hygiene practice and risk factors of street food vendors. Huynh-Van et al. (2022), entitled "Factors associated with food safety compliance among street food vendors in Can Tho city, Vietnam: implications for intervention activity design and implementation," studied the factors associated with compliance of food safety among street food vendors in Vietnam. Dollentas & Habla (2021), "Food Hygiene and Safety

Practices of Food Establishments," studied food hygiene and safety practices of food establishments. Limon (2021), "Food safety practices of food handlers at home involved in online food businesses during the COVID-19 pandemic in the Philippines," examined food safety practices of food handlers at home involved in online food businesses during the COVID-19 pandemic in the Philippines. Origenes et al. (2022), "Microbiological Load of Street Food and To What Extent the Food To Safety To Practices are Complied with by the Vendors in the Schools of Cebu City," analyzed the microbiological load and compliance of vendors' food hygiene and safety practices in Cebu City, Philippines. Caye (2024), in "Food Safety Awareness and Practices of Street Food Vendors in Selected Barangays, Marawi City," measured food safety awareness and practices of street food vendors in Marawi City, Philippines. Jose & Villanueva (2023), in "Evaluation of the Hygiene Procedures and Commitment to Food Safety of Calapan City Street Food Vendors," evaluated the street food vendors' hygiene procedures and commitment to food safety in Calapan City, Philippines. Rifat et al. (2022), in "Food safety knowledge and practices among food handlers in Bangladesh: A systematic review," systematically reviewed food handlers' food safety practice and knowledge in Bangladesh. Moges et al. (2023), in "Sanitary Condition and Hygienic Practice of Street Food Vendors in Selected Towns of Ethiopia: A Cross-Sectional Study Addressing Public Health Concern," evaluated the sanitary condition and hygienic practice of street food vendors of selected towns in Ethiopia. Milovanova et al. (2024), "Understanding food safety challenges through a gender and social lens in low- and middle-income countries," addressed food safety challenges from a social and gender perspective in low- and middle-income countries. Research repeatedly identifies a range of determinants of food safety practice. Knowledge and education are the drivers. Positive attitudes towards food safety also matter as positive-attitude food handlers are more probable to adopt safe

habits. Socioeconomic status in the form of income, education, and residence matters, and studies in Bangladesh and other nations have shown that higher income and better residence are associated with better food safety practice and knowledge. As per the research, various interventions and suggestions are feasible to improve food safety. Special training courses are required for food handlers, giving priority to priority topics such as personal hygiene, avoiding cross-contamination, and time and temperature control. Public education schemes can educate consumers and food handlers about food safety risks and safe practices. Upgrading infrastructure, including sanitation facilities, water supply, and waste management services, is required, particularly in poor environments. Policy and regulatory measures need to be upgraded for obtaining compliance of food vendors and food premises. Community-based interventions would enhance food safety knowledge and practice at the household level. The majority of research studies carried out on food safety practices, attitudes, and knowledge affirm the need to address this critical public health concern.Foodborne illness cases would be reduced and public health as well as wellbeing would be improved through implementation of some interventions and intervention of food safety practice determinants. Literature review of food safety knowledge, attitudes, and practices discusses a set of converging food safety determinants in different contexts. From developing-country street vendors through restaurants to homes, there is a requirement for greater food safety controls. Adopting evidence-based therapies, championing education and illumination, and narrowing socioeconomic divides, we can aim to inhabit a less foodborne-disease-prone world with an improved public health.

Age

Age is a significant demographic factor in the food safety practices of street food vendors, with its impact expressed differently across age groups and geographical

locations. Werkneh et al. (2023) hypothesize that middle-age vendors will be likely to adhere scrupulously to formalized food safety regulations existing in the market, a factor traceable to the build-up of experiential learning and consequent accent on the maintenance of rigid hygienic standards. By contrast, there is a call in the existing literature for heterogeneity in age in the susceptibility of vendors to formalized food safety training programs (Sandri, 2024). Specifically, young sellers show a greater inclination towards the uptake of the content of such programs, while their older counterparts may encounter significant inhibitive factors in implementing new operational procedures. This calls for the implementation of customized pedagogical practices based on the rigorous assessment of the cognitive capacity inherent within each particular age group. Solon's (2022) empirical studies ("Food Safety Practices among Street Food Vendors in the Twin Cities of Zamboanga Del Norte") concluded that vendors past the age of forty years in the Zamboanga del Norte region exhibited a demonstrably higher level of compliance consistency with recommended hygienic practices. Likewise, Nortey et al. (2024) identified increased compliance levels among vendors aged 25-40 years old working in the Takoradi Sub-Metropolis of Ghana. Nevertheless, it is relevant to consider the cross-cutting evidence by Negassa (2023), and that was drawn on lower compliance levels among Ethiopian vendors aged 26-35 years old, thus pointing to age effects being local context specific. Nkosi et al. (2021) reported that older Zululand vendors had wide knowledge of food safety customs, but this did not automatically equate to enhanced practice. Desye et al. (2023) found that young vendors have strong beliefs but weak practice, while older vendors apply experiential learning but are resistant to embracing new safety practices without guided training. Dhudum et al. (2025) and Abu-Almaaly (2025) have also contributed to this body of knowledge. There is additional research on this interaction by Parida et al.

(2025), Moges et al. (2024), Carpio (2025), Rosales et al. (2023), and Yakubu et al. (2023). There is emphasis on the worldwide importance of food safety by the WHO/FAO (2022).

The intricate interaction between in order age and the adherence to food safety practices among ambulatory vendors additionally relies upon a range of interlinked socio-economic variables. The investigation of Rahman et al. (2022), conducted within the geographical limits of Dhaka, Bangladesh, revealed a statistically significant correlation between vendors' formal educational attainment, availability of capital, and the adherence with which they adhered to established food safety protocols. This observation suggests that chronological age, though indisputably relevant, becomes irrevocably rooted in a wider universe of determinants encompassing but not limited to formal schooling, buying power, and access to key resources. Moreover, one must also acknowledge the overriding influence of culturally-prescribed practices and deep-rooted culinary traditions on the determinants of food handling behavior seen among these vendors. It is common to find that older vendors possess an inventory of customary food knowledge pertaining to preservation and preparation techniques; however, the epistemological integrity of these practices may not always parallel existing evidence-based requirements for food safety. Because of this, there is an inherent need for the development and implementation of culturally-sensitive intervention steps, designed to adequately balance practice based on tradition with strict adherence to contemporary food safety procedures. The work of Jallow et al. (2023) in The Gambia indicated that food traders operating within jurisdictions featuring severe regulatory control and frequent inspections displayed greater adherence to food safety legislation, thereby emphasizing the role of effective regulatory systems and effective enforcement mechanisms. Consumer culture and demand for secure food are also factors. Nguyen et

al.'s (2024) research in Vietnam found that vendors serving health-aware consumers had a greater likelihood to focus on food safety and hence the role of consumer campaigns and awareness was highlighted. In conclusion, although age is a significant factor, a combination of socio-economic, cultural, regulatory, and consumer-related determinants altogether influences food safety behavior among street vendors. Intervention should be multi-determined and holistic in nature to counter these multi-determination influences as well as ensure safety and well-being for both consumers and vendors.

Studies are important to reveal the formation of food safety behavior during the entire life and with increasing experience. These studies can provide useful information about the long-term effects of training interventions and regulatory interventions on vendor behavior. Furthermore, qualitative research instruments, such as in-depth interviews and ethnographic studies, can provide intensive insight into the food safety behavior influenced by the cultural and social factors. By combining the quantitative and qualitative approaches, researchers will have the ability to gain a better understanding of the intricate dynamics of the food safety driving variables among street food vendors. Subsequent research should therefore focus on the study of intersectionality among these variables as well as designing specific strategies for enhancing food safety among this vulnerable population. To further purify our knowledge, subsequent academic studies should then attempt to adopt a multi-methodological strategy that combines quantitative analysis with qualitative research in an attempt to capture the complex dynamics of food safety attitudes and practices in various socio-cultural settings. This would allow for the formulation of evidence-based intervention programs, depending on the idiosyncratic requirements and issues confronting street food vendors within heterogeneous demographic strata. A

wide range of research articles has also placed emphasis on the multi-faceted nature of food safety practices among street food vendors.

According to Solon (2022) ("Food Safety Practices among Street Food Vendors in the Twin Cities of Zamboanga Del Norte") provided empirical evidence of the Twin Cities of Zamboanga Del Norte on regional variation in following food safety practices. Ndhlovu (2024) ("An Investigation of the Food Safety Knowledge of Street Food Vendors and Sanitary Conditions of Their Vending Sites") added greatly to street food vendors' knowledge and sanitary conditions. Ibrahim and Adeola (2024) ("ASSESSING FOOD SAFETY AWARENESS AND PRACTICES AMONG STREET FOOD VENDORS IN KATSINA STATE, NORTH WESTERN NIGERIA") conducted an assessment of food safety awareness and practices in Katsina State, North Western Nigeria, providing a regional perspective on the challenges and opportunities in this sector. Imam (2024) ("Knowledge on Food Safety and Hygiene Practice among Street Food Vendors of Dhaka City") handled the knowledge and practice of hygiene among street food vendors in Dhaka City and diagnosed the specific issues within this urban setting. Yi and Choi (2025) ("Age differences in food safety knowledge, attitudes, and practices among consumers in Seoul") explored age differences between consumers' food safety knowledge in Seoul and presented a more contextualized perspective on food safety attitudes. Gebre (2023) ("Food safety knowledge, attitude, and practice among male and female food handlers: Evidence from fruit and vegetable producers in Ethiopia") examined the food handling practice and knowledge of food handlers in Ethiopia, while Garsow et al. (2022) ("Exploring the relationship between gender and food safety risks in the dairy value chain in Ethiopia") examined the link between food safety risks and gender in the Ethiopian dairy value chain. Zanin et al. (2021) ("Influence of educational actions on transitioning of food safety (FS)-culture: a longitudinal study using action research in an

army food service setting") investigated the impact of educational interventions on food safety culture, and Girmay (2020) ("Determinants of Sanitation and Hygiene Status Among Food Establishments in Addis Ababa, Ethiopia") determined food establishments' sanitation and hygiene determinants in Addis Ababa. Agrawal et al. (2025) ("Street Food Safety: A Study on Hygiene Practices and Risk Factors among Vendors") researched hygiene practices, Huynh-Van et al. (2022) ("Factors associated with food safety compliance among street food vendors in Can Tho city, Vietnam: implications for intervention activity design and implementation") researched compliance in Vietnam, and Dollentas and Habla (2021) ("Food Hygiene and Safety Practices of Food Establishments") researched food establishments. Origenes et al. (2022) ("Street Food's Microbiological Load and Vendors' Food Hygiene and Safety Practices Compliance in the Schools of Cebu City") examined Cebu City street food, Jose and Villanueva (2023) ("Evaluation of The Hygiene Procedures and Commitment to Food Safety of Calapan City Street Food Vendors") examined Calapan City vendors, Siddiky et al. (2024) ("Determinants of Food Safety Knowledge and Practices among Food Handlers in Bangladesh: An Institution-Based Cross-Sectional Study") examined Bangladeshi food handlers, Rifat et al. (2022) ("Food safety knowledge and practices among food handlers in Bangladesh: A systematic review") conducted a systematic review, and Aljasir (2023) ("Food safety knowledge and practices among food handlers and consumers in Gulf countries: An integrative review") reviewed the Gulf countries' food safety. The studies as a whole point toward the need for evidence-based and context-dependent interventions to improve the food safety standards of street food vendors.

Sex

The effect of gender on hygiene behavior among street food vendors, and the resultant implications for safety, is a complex issue with worldwide varied and

context-dependent results. Despite the fact that a majority of street food sellers are women, their hygiene practice may not always resemble that of men but according to most research, gender would not be a factor in itself for differentiating variation in sanitation practices. In response to effective management of food, the Food and Agriculture Organization (FAO) and World Health Organization (WHO) recommend gender-sensitive approaches in food safety intervention and training. Koumassa (2025) is convinced that women vendors constituting more than 55% of employees in most cases may be disadvantaged by the obstacles posed by limited access to sanitation facilities and education, which may compromise their following best hygiene standards. Yet, Begum (2024) discovered that differences in food safety behaviors in Bangladesh were not significantly related to street food vendors' gender. In the same vein, Solon (2022) found that gender did not predict variation in sanitation among street food vendors operating within the Twin Cities of Zamboanga del Norte, while Carpio (2025) concluded that there was no variation in food safety and hygiene practices overall between male and female vendors in Alfonso Lista, Ifugao. In Nigeria and the rest of Africa, cultural-based gender roles can affect hygiene behavior; according to Kaddouri (2025), female vendors would be more meticulous when it comes to clothing and cleanliness of equipment due to roles at home, whereas male vendors would be less meticulous about time-consuming activities such as disposal of garbage. Girma (2024), Klutse (2025), and Girma (2025) favor context-specific approaches to intervene in the specific challenges and opportunities of male and female vendors to promote safer food handling practices. Ndhlovu (2024) examined food safety knowledge among street food vendors, while Ibrahim (2024) and Adeola (2024) evaluated food safety awareness and practice among street food vendors in Katsina State, Nigeria. Imam (2024) investigated food safety and hygiene behaviors of street food vendors in Dhaka City. Huynh-Van et al.

(2022) explored factors related to food safety compliance among street food vendors in Can Tho city, Vietnam. Agrawal et al. (2025) examined hygiene practice and risk factors among vendors, Dollentas and Habla (2021) determined food hygiene and safety practice of food businesses, and Origenes et al. (2022) analyzed the microbiological load of street foods and compliance of vendors in Cebu City schools. Jose and Villanueva (2023) assessed hygiene protocols and food safety commitment among Calapan City street food vendors. Siddiky et al. (2024) investigated food safety knowledge and practice determinants among food handlers in Bangladesh, Rifat et al. (2022) systematically reviewed food safety knowledge and food handlers' practices in Bangladesh, and Aljasir (2023) presented an integrative review of food safety knowledge and food handlers' and consumers' practices in Gulf countries. Gebre (2023) showed data from Ethiopian fruit and vegetable producers, Garsow et al. (2022) investigated gender-food safety risks linkage in the dairy value chain in Ethiopia, and Girma (2020) analyzed sanitation and hygiene status determinants in food establishments of Addis Ababa, Ethiopia. Zanin et al. (2021) investigated the effect of educational behaviors on the transfer of food safety culture in a military food service environment, and Yi and Choi (2025) examined age differences in food safety knowledge, attitudes, and consumer practices in Seoul. The gender and hygiene practice among street vendors discourse is more than just mere conformity to developed procedures and delves into socio-economic and cultural determinants of individual behaviors. Availability of resources such as clean water, access to sanitation facilities, and access to waste disposal means, usually decide to what extent vendors can keep high hygiene standards. These resources may not necessarily be readily available in developing nations, particularly in unofficial vending spots, placing a disproportionate burden on female vendors who are already disadvantaged in society.

Also, cultural norms and expectations can influence the type of foods which are being sold, the way foods are prepared, and the general hygiene strategy. In some cultures, women have significant roles to play in preparing food and keeping homes clean, and this can be translated to a greater awareness level on the practice of hygiene within the vending environment. On the other hand, in societies where men control the street food sector, various levels of hygiene may be practiced, perhaps dictated by aspects like physical power needed for specific operations or customary notions of cleanliness. Training and education have a significant function to foster safe food handling behavior among street food vendors, be it male or female. The effectiveness of such interventions is determined by whether or not they are useful in responding to the particular needs and constraints of vendors within various cultural contexts. Training that is not culturally appropriate or that does not respond to vendors' practical limitations may not be effective in shifting behavior.

In addition, consumer attitudes and expectations may also determine the hygiene behavior of street food vendors. If consumers place more importance on taste and price than on hygiene, then vendors will have less incentive to expend resources on expensive hygiene. Thus, consumer education on food safety and hygiene plays a vital role in establishing demand for safer street food. The involvement of regulatory bodies and the government in maintaining food safety in street foods cannot be overemphasized. Enforcing food safety standards effectively, in addition to measures in increasing resource availability and training, can make the environment more favorable for vendors to practice and sustain good hygiene control. Regulations must, however, be framed in an equitable and fair manner so as not to have unintended effects likely to adversely impact female vendors or other vulnerable groups. The gender dynamic and the practice of hygiene among street food vendors is a multifaceted and

context-dependent phenomenon that requires understanding social, economic, and cultural contexts of the vendors. Food safety intervention has to be region-specific and problem-oriented and grounded in needs and challenges of vendors in different areas with sensitivity to factors such as resource accessibility, cultural setting, educational levels, and consumer culture. Through an integrated and context-specific approach, we are able to create a safer and sustainable street food business for the benefit of vendors and consumers.

Educational Attainment

The educational attainment and food safety behavior of street food vendors have a complex interrelation that sometimes produces contradictory findings between different studies. For example, Solon (2022), in "Food Safety Practices Among Street Food Vendors in the Twin Cities of Zamboanga del Norte," found that most street food vendors in the area held college degrees but that a majority (98%) of them were not trained formally in food safety. What this suggests is that formal education alone is perhaps not sufficient to ensure good hygiene practices are being followed. This corresponds with Tamiru (2022), who studied food handlers working in food outlets in Southwest Ethiopia, further underpinning the notion that education and applied food safety awareness are not necessarily correlated. Abid (2022) reported a divergent trend in Chattogram, Bangladesh, wherein high educational achievements were positively related to greater numbers of street food vendors following better food safety practices. This finding is supported by Chopra's (2021) systematic review, whereby he highlights the broader role of training and education to enhance food safety performance. Ethiopian research evidence is reserved in speaking about this complex dynamic. Chekol (2021), Adem (2023), Negassa (2023), Moges (2023), Azanaw (2022), and Werkneh et al. (2023) all examine different aspects of street food vendors' food safety and hygiene and

make evidence-based evidence on the subject from a wide angle. Moges, Rødland, and Argaw (2023) in their "Sanitary Condition and Hygienic Practice of Street Food Vendors in Selected Towns of Ethiopia: A Cross-Sectional Study Addressing Public Health Concern" write about public health issues regarding street food vendors in Ethiopia.

More studies from across the globe contribute to this insight. Hassan (2020) assessed Zanzibar's food safety behavior, whereas Wallace (2022) carried out a review of vendor knowledge, attitudes, and practices in low- and middle-income contexts. Nortey (2024) in Ghana, Islam (2024) in Dhaka ("Inadequate Food Safety Knowledge and Hygiene Practices Among Street Food Vendors in Dhaka, Bangladesh" by Islam, Tanjia, Mitra, et al., 2024), and Gonzales (2023) in the Philippines also provide helpful information within this context. Systematic reviews by Cataluna (2024) and Desye (2023) provide wider summaries of food hygiene and safety among street food vendors. Desye, Tesfaye, Daba, and Berihun (2023) performed a systematic review and meta-analysis of "Food safety knowledge, attitude, and practice of street food vendors and associated factors in low- and middle-income countries". Additional context comes from research in other places, such as Ghosh (2023) in Kolkata, Addo-tham (2020) in Ghana, Madilo (2022) among school food handlers, Girma (2024) in Ghana, and Wu (2024) in China. Various other studies shed more light on food safety knowledge, food safety practices, and the related factors among street food vendors. Ndhlovu and Tabit (2024) analyzed food safety knowledge and sanitary conditions, and Ibrahim (2024) and Adeola (2024) measured food safety awareness and food safety practices in Nigeria. Imam (2024) looked at hygiene practices in Dhaka, and Yi and Choi (2025) studied age differences in Seoul. Gebre (2023) presented data from Ethiopian fruit and vegetable farmers, while Garsow et al. (2022) explored gender-specific risks for the dairy sector in Ethiopia. Zanin et al. (2021) researched educational activities' impact on food safety culture, and Girmay

(2020) looked at determinants of sanitation in Addis Ababa. Agrawal et al. (2025) reported on hygiene practices and risk factors, whereas Huynh-Van et al. (2022) investigated factors related to food safety compliance in Vietnam. Dollentas and Habla (2021) investigated food hygiene practices, and Origenes et al. (2022) investigated microbiological load and hygiene compliance in Cebu City. Jose and Villanueva (2023) assessed hygiene protocols in Calapan City, Siddiky et al. (2024) established determinants of food safety information in Bangladesh, and Rifat et al. (2022) systemically reviewed food safety information in Bangladesh. Aljasir (2023) conducted an integrative review of food safety knowledge in Gulf states, Milovanova et al. (2024) emphasized gender and social perspectives in low- and middle-income nations, Chen et al. (2025) investigated drivers of good food-handling practices, Borbon & Tolentino (2020) wrote about food safety practices for policy improvement, Limon (2021) investigated food safety practices amid the COVID-19 pandemic in the Philippines, and Caye (2024) investigated food safety awareness in Marawi City. A number of non-educational factors affect food safety behavior. Huynh-Van et al. (2022) concluded that education, food safety training, business capital, and vending area size were positively related to food safety compliance by street food vendors in Vietnam. The mobile vendors were also less likely to comply with food safety standards. Basheikh, Jumbe, and Kulwa (2023) found that attitudes and beliefs of vendors towards healthy meals are related to sex, education level, and investment cost. These disparate findings provide the rationale for holistic interventions in promoting food safety behaviors among street food vendors. Education is necessary but effective only sometimes. Training programs, infrastructure enhancement, and facilitative policies are also required to make sure that street food vendors learn and maintain safe food handling techniques. In the future, future studies need to emphasize how to find the most

effective methods to encourage food safety among street food vendors across various environments. This involves looking into innovative ways to provide food safety training, enhancing access to resources and facilities, and enhancing regulatory enforcement. In addition, longitudinal studies would be required to determine the long-term effects of interventions and determine the factors associated with long-term behavior change. Through a greater evidence base, researchers will be able to guide intervention development that is targeted and effective in defending public health and upholding the livelihoods of street food vendors.

This connection between educational level and food safety practice among street vendors is complex and not always linear. Education may improve knowledge, but its effect on actual actions differs greatly within varying contexts. The access to formal food safety training, the availability of resources, and the effectiveness of regulation all contribute significantly to the determination of vendors' actions. Evidence-based holistic interventions integrating education with on-the-job training, investment in infrastructure, and pro-poor policies are needed to foster safer food handling practices. Future studies should aim at establishing the most efficient means of fostering sustained behavior change among street food vendors in order to ultimately protect public health and ensure the livelihoods of these critical community members.

Registered Vendors

Recent literature on food safety practices among street food vendors in the Philippines identifies licensing as a significant factor, hence providing the context in which the situation in Buenavista and Nasipit should be considered. Armas et al. (2024) emphasize a strategic approach to mobile food safety in the International Review of Management & Marketing and indicate that licensing may serve as a means of assuring

adherence to standards of safety. Food safety knowledge and practices, which licensing can supportively reinforce, were emphasized in studies from Alfonso Lista, Ifugao (Carpio et al., 2025), Estancia, Iloilo (Belgica, 2024), and the Fourth District of Iloilo (Bermejo, 2024). These respective findings suggest that licensing can be used as a basis for the review and improvement of policies toward safer street food vending in the Philippines.

Other related studies in Biliran Province by Gonzales & Pabilando (2023) and in Calapan City by Jose & Villanueva (2023) provide further details on how licensing bears implications for hygiene practices and methods of food handling. These studies have shown that through licensing, specific standards can be upheld and practiced by vendors. This is a means of verifying hygiene practices, as well as the concern for food safety. These studies enhance the perspectives on how licensing can affect the quality of street food vending.

Finally, and perhaps most importantly, the literature underlines that licensing is an important aspect of food safety and not simply a regulatory requirement for street food vendors. With its provisions for policy review, enforcement of hygiene standards, and enhancement of food safety knowledge, licensing could fulfill an essential function in protecting public health in districts such as Buenavista and Nasipit.

Unregistered Vendors

Recent studies on food safety practices among street food vendors in the Philippines introduce a relevant backdrop for research into the specific context of Buenavista and Nasipit, with a focus on non-licensing aspects. For example, research in Aklan (Paloma & Israel, 2025) points to microbial counts and safety practices that have important implications for health risks. Indeed, in Marawi City, Caye (2024) and in Catarman,

Northern Samar, Madario et al. (2023) underline the need for raising the levels of awareness about food safety and good manufacturing practices. These cumulative findings support the importance of comprehensive assessment and targeted intervention in street food vending to improve food safety in areas like Buenavista and Nasipit, where non-licensing factors will play a crucial role.

Analysis reveals the multidimensional nature of food safety. Tayco et al. (2025) assess the knowledge in food safety, attitude, and practices of vendors to provide an overall perspective on their behaviors. Cabatingan et al. (2025) discuss chosen street food vendors in relation to these areas, adding to the broader understanding of challenges and opportunities for improvement. In focusing on awareness, practices, and attitudes, these studies underscore how a holistic approach to ensuring food safety among street food vendors is highly relevant in regions like Buenavista and Nasipit.

1.4 Schematic Diagram

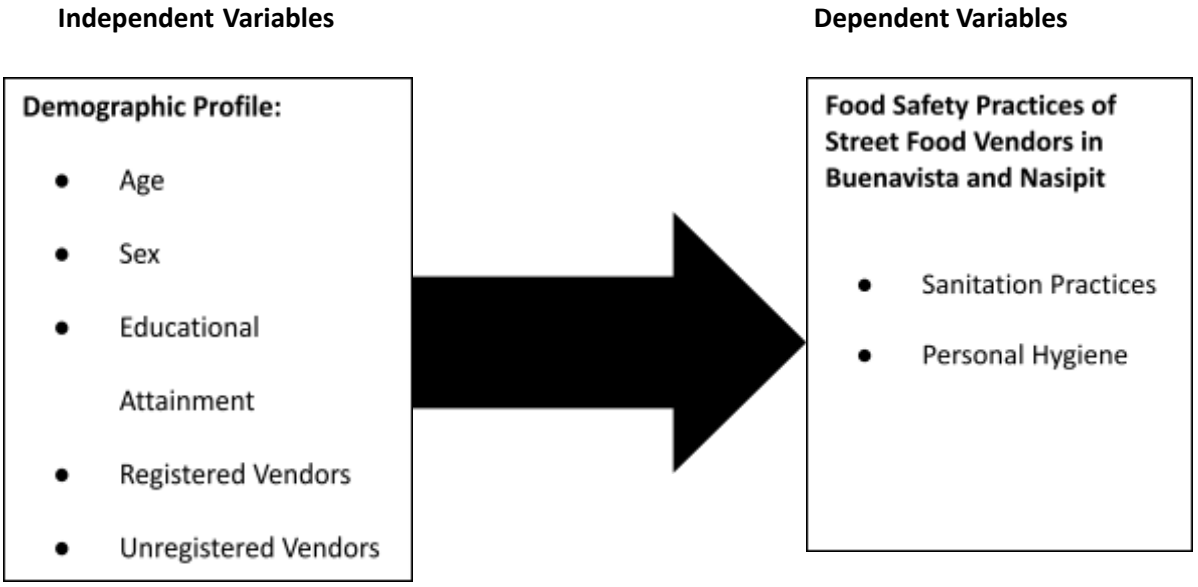


Figure 1. It shows how respondents' demographic characteristics may affect their personal hygiene and sanitation practices.

1.5 Statement of the Problem

This study will assess the food safety practices among street food vendors in Buenavista and Nasipit, Agusan del Norte.

Specifically, the study tried to answer the following questions.

1. What is the profile of the respondents in terms of:

1.1 Age;

1.2 Sex, and

1.3 Educational Attainment,

1.4 Registered Vendors,

1.5 Unregistered Vendors;

2. What is the level of food safety practices as perceived by the street food vendors and customers in terms of:

2.1 Sanitation Practices; and

2.2 Personal Hygiene;

3. Is there a significant relationship between the respondents' profiles and their perceptions of food safety practices among street food vendors in Buenavista and Nasipit?

1.6 Hypothesis

There is no significant relationship between the respondents' profiles, which include age, sex, educational attainment, and their level of food safety practices in sanitation and personal hygiene. On the other hand, the alternative hypothesis is that a significant relationship exists between these variables.

1.7 Scope and Limitation

This research assesses the food safety practices of street food vendors in Buenavista and Nasipit, Agusan del Norte. It intends to identify vendors' knowledge,

attitude, and practice regarding food handling, hygiene, storage, and sanitation. This study will specifically cover vendors who sell ready-to-eat food products within public markets, streets, and commercial areas, and both licensed and non-licensed food stalls in the said municipalities. Information will also be collected using surveys, interviews, and observational checklists to determine their overall food safety practices.

However, this study is limited to the vendors' practices and perceptions. It does not involve direct microbiological testing of the food that they sell. The research does not cover food vendors outside the municipalities under study or selling pre-packaged or pre-prepared foods. This study also relies on self-reported feedback and observations collected at a given time. It can fail to capture all changes in practice over time. Weather conditions or seasonal changes impacting food safety are environmental factors out of the scope of this study.

Lastly, although this study seeks to establish gaps in food safety practice and suggest potential interventions, it does not discuss local government units' enforcement or regulation policies in detail.

1.8 Definition of Terms

To better comprehend the study, the following terms are operationally defined:

Age. Refers to the age in years that a vendor and customer have lived, which can influence food safety knowledge.

Sex. Refers to the biological sex of the vendor and customer, whether male or female.

Educational Attainment. Refers to the highest education level a vendor and customer have achieved.

Sanitation Practices. Refers to the methods vendors apply for cleaning purposes to ensure food safety.

Personal Hygiene. Refers to the habits vendors have for personal cleanliness to avoid food contamination.

Convenient Food. Refers to Pre-cooked or easily prepared food that the vendor is selling.

Small-scale Businessperson. Refers to a vendor running a small street food stall business.

Foodborne Disease. It refers to sickness from consuming openly prepared foods and not taking sanitation precautions.

Peddlers. Refers to a street food vendor as they travel or change location to sell their products.

Food Safety Practices. Refers to the practices vendors undertake to make food safe for consumption.

Ambulant Vendors. Refers to street sellers with no fixed location of sale.

Imperative Intervention. Refers to the necessary practices to enhance food safety.

Evidence-based. Refers to real data or research findings.

Planned Behavior. It refers to the idea that intention and attitude drive behavior.

Vendor. Refers to the person who sells ready to eat foods in the market and outside the school campus.

Street Food. Refers to the convenient and affordable ready to eat foods.

CHAPTER 2

RESEARCH METHODOLOGY

2.1 Research Design

This research will utilize a descriptive cross-sectional quantitative research design to assess food safety attitudes, knowledge, and practices (KAP) among food vendors in Nasipit and Buenavista, Agusan del Norte. Based on the Extended Theory of Planned Behavior (TPB) and the KAP model, this research seeks to investigate behavioral intentions and determine areas of gaps between vendors' knowledge, beliefs, and practices on food hygiene and sanitation. The study population is registered vendors in public markets, transport zones, and school premises. The researchers will recruit a proportionate stratified random sample of 15–20 vendors from the two towns. Data will be collected using a structured, validated questionnaire to assess food safety knowledge, attitudes, practices, and TPB constructs, as well as a standardized observation checklist to assess actual hygiene activities and environmental status. The researchers will explore the quantitative data using descriptive statistics (means, percentages, frequency) and inferential statistics such as chi-square tests, t-tests/ANOVA, and correlation or regression analysis to identify relationships between variables. Ethical issues include informed consent, confidentiality, and ethics board approval. The study consists of only two towns and does not seek to test seasonality or nutritional content. The researchers expect the results to provide baseline data on vendor behavior, identify risky behavior, and inform evidence-based policy and training interventions to enhance street food safety in the area.

2.2 Research Materials and Methods

This research will utilize a descriptive cross-sectional quantitative research design to assess food safety attitudes, knowledge, and practices (KAP) among food

vendors in Nasipit and Buenavista, Agusan del Norte. Based on the Extended Theory of Planned Behavior (TPB) and the KAP model, this research seeks to investigate behavioral intentions and determine areas of gaps between vendors' knowledge, beliefs, and practices on food hygiene and sanitation. The study population is registered vendors in public markets, transport zones, and school premises. The researchers will recruit a proportionate stratified random sample of 15–20 vendors from the two towns. The researchers will gather data through a guided, validated questionnaire to measure food safety knowledge, attitudes, practices, and TPB constructs, as well as a standard observation checklist to assess actual hygiene practices and environmental conditions. The researchers will analyze the quantitative data using descriptive statistics (means, percentages, frequency) and inferential statistics such as chi-square tests, t-tests/ANOVA, and correlation or regression analysis to identify relationships among variables. Ethical concerns are informed consent, confidentiality, and ethics committee approval. The research only uses two towns and does not attempt to test seasonality or food content. The researchers expect the outcomes to provide baseline information on vendor actions, assess risky behaviors, and support evidence-based training and policy interventions to improve regional street food safety. Vendor food safety gaps include low knowledge, poor hygiene, poor storage, and poor temperature control: insufficient health checks, pest control, and labeling increase foodborne illness hazards. Education, sanitation, and stronger management can improve practices and protect public health.

In presenting, analyzing, and interpreting data, the researcher used the following statistical test, namely.

Frequency and Percentage —The researchers utilized this instrument to characterize the participants' demographic profile, sex, age, educational level, and years of vending or frequency of buying street food.

Mean - This instrument measured street food vendors' and consumers' food safety behavior in Buenavista and Nasipit.

Correlational Analysis - The researchers used this instrument to collect and analyze the respondents' demographic profiles and evaluate their level of food safety practice.

2.3 Research Environment, Population, and Sample

The researchers will conduct the study in the Municipalities of Buenavista and Nasipit, which have an emergent street-food culture where vendors serve multiple foods and drinks in public areas such as terminals, schools, markets, and parks. Street food selling in these localities offers affordable food to the public. It is a significant source of livelihood for most people. However, it also has the potential to cause food safety and sanitation concerns. Hence, the environments are perfect for judging vendors' food preparation, handling, and selling practices and identifying the determinants of their compliance or lack thereof with food safety regulations. The study population will comprise all street food vendors operating in public places in Buenavista and Nasipit, especially those selling cooked or ready-to-eat foods like grilled meat, fried foods, drinks, and other local specialties. The researchers will include fixed vendors with permanent stalls and mobile or ambulant vendors who move from place to place. Researchers can utilize local government documents or business licenses, whenever available, to estimate the number of vendors in each area. Food handling and preparation participants will be selected using a purposive sampling technique. The sample will rely on the number of available vendors, and the researchers will aim to include 15–20 vendors from each municipality to facilitate comparison. The researchers will select respondents on a voluntary and convenience sampling basis during the data collection time, and they will interview them or ask them to complete questionnaires to assess their food safety knowledge, attitudes, and practices. Vendor selection will be

based on responsiveness while guaranteeing sample representativeness. The data collection procedure will be courteous and easy for vendors, with unforgiving confidentiality and anonymity measures. Studies in such environments will yield significant evidence regarding street food vendors' food safety practices, to guide targeted interventions to a safer food environment. The findings of this study can guide policy at both national and local levels to underpin Sustainable Development Goals in food security, health, and well-being.

The respondents of this survey of street food vendors' food safety habits are vendors and consumers from the Buenavista and Nasipit municipalities. Each area will have approximately 15 to 20 street food vendors and 15 to 20 consumers, 30 to 40 respondents per municipality. Purposive sampling will be employed to pick vendors currently selling in congested areas like markets and terminals. In contrast, the researchers will use convenience sampling on available and willing-to-be-surveyed customers. Vendors and consumers will paint a wider picture of street food safety among the two communities.

Table 1. Distribution of Respondents

Municipality	Respondent Type	Estimated Total Population	Target Respondents	Sampling Method
Buenavista	Street Food Vendors	29	15-20	Purposive Sampling
Nasipit	Street Food Vendors	21	15-20	Convenience Sampling
Buenavista	Consumers	15-20	15-20	Purposive Sampling
Nasipit	Consumers	15-20	15-20	Convenience Sampling
Total			60-80	

Table 1. This table will present a total of 15-20 vendors and 15-20 consumers from Buenavista and Nasipit will be surveyed in this study. Purposive sampling will be used for selecting vendors in Buenavista, while convenience sampling will be used for Nasipit. There were also 15 to 20 target consumers in each municipality, with a similar method of sampling to be used as with the vendors. The total respondent pool was around 60-80 participants.

2.4 Research Data and Analysis Plan

This research employed a researcher-altered questionnaire checklist based on the instrument Solon (2022) utilized in the study titled "Food Safety Practices among Street Food Vendors in the Twin Cities of Zamboanga del Norte." The researchers selected the instrument because it applies to this research, as both studies measure the food safety practices of clients and street food vendors in comparable Philippine locales.

The researchers divided the survey into two sections. Part I collected demographic data from the respondents, such as gender, age, education level, and frequency of buying street food. Part II aimed to measure food safety practices, particularly personal hygiene and street food vendors' sanitation practices. The researchers constructed the variables in this aspect to examine how the suppliers treat food, maintain personal hygiene, and meet hygiene standards while selling.

The tool utilized a four-point Likert scale with answer options from 1 (Strongly Disagree) to 4 (Strongly Agree) to assess the prevalence of food safety practices. Minor changes were incorporated when translating the questionnaire into the Buenavista and Nasipit environments. The tool was pre-tested for expert appraisal before administration to confirm its understandability, content validity, and local condition appropriateness.

Data Gathering Procedure

The information for this study will be collected from street food vendors and consumers in Buenavista and Nasipit to evaluate their food safety practices on the aspects of sanitation and personal hygiene. The researchers will employ a structured questionnaire with a 4-point Likert scale and an observation checklist as the primary data collection tools. Prior to data collection, the tools will be validated by the adviser and experts, and clearance to undertake the study will be obtained from Saint Michael College of Caraga as well as local authorities. Respondents will be chosen using purposive and convenience sampling, namely 15 to 20 vendors and 15 to 20 consumers from each municipality. The researchers will conduct the questionnaires personally and observe vendors' sanitation and hygiene practices using the checklist, including hand washing, food storage, and the cleanliness of the vending area. Participants will be told the purpose of the study, assured of confidentiality, and asked for voluntary consent. The gathered data will be verified every day, encoded, and analyzed with frequency, percentage, mean, and correlational analysis while maintaining confidentiality and ethical standards.

2.5 Research Ethical Standards

A. Intellectual Property Rights

This study will respect intellectual property rights by maintaining respondent anonymity throughout the analysis. The researchers will correctly cite all secondary data and existing literature, honoring the intellectual property of others while protecting the researcher's contributions.

B. Informed Consent

The research's respondents will be the street food vendors of Buenavista and Nasipit, Agusan del Norte. Before their involvement, the researchers will conduct a

short orientation to articulate the purpose of the study, which will focus on validating their food safety practices. The orientation will also highlight the potential benefits and negligible risks involved. Respondents will be informed that their involvement is entirely voluntary and that they may withdraw from the study without giving any reason. To ensure understanding and assent, the researchers will distribute a written consent form to each respondent during the orientation and will clearly explain the respondents' rights and the purpose of the study.

C. Data Privacy and Confidentiality

The researchers will collect personal data, record respondents' demographic profiles, handle the data strictly, and obtain respondents' explicit consent through an informed consent form. All data will be anonymized to protect respondents' identities. The questionnaires will be administered under the researchers' supervision to ensure respondents' confidentiality. Only the researchers will have access to the code linking identification numbers to the respondents. All data will be collected and handled with the highest level of confidentiality, and the researchers will fully protect respondents' privacy throughout the research process.

D. Voluntary Participation

Participation in the study will be voluntary. The researchers will inform the respondents of their right to withdraw without negative consequences. No animals will be involved in this study; therefore, considering alternatives to animal use will not be applicable.

E. Minimization of Harm

The researchers will engage with the respondents respectfully and professionally to establish a safe and supportive environment throughout the study. If any risks are associated, respondents will be referred to the appropriate support

services to ensure their well-being, such as the clinic and the Guidance Office for assistance if needed during the research process. After participation, researchers will provide a debriefing session to ensure no physical, psychological, or emotional harm occurrs throughout the research process. The study will not involve animal subjects; the 3Rs replacement, reduction, and refinement principles will have no application.

F. Beneficence

This research will help deepen the knowledge about the street food vendors' food safety practices in Buenavista and Nasipit and the challenges they encounter to ensure hygiene and sanitation. The survey results will be made available to respondents if requested. The benefits of the study are to advance local food safety standards, inform public health programs, and enhance policy-making to ensure safer street food vending.

G. Justice and Fair Participant Selection

Respondents will be selected based on well-defined and fair inclusion criteria, ensuring the selection process aligns with the study's objectives. Respondents from diverse backgrounds will be included fairly without discrimination.

H. Data Integrity and Accuracy

The researchers will collect and analyze data using standardized methods to ensure accurate and reliable results. Initially, they submit a formal request to the school registrar to obtain the complete list of eligible respondents for the study. All findings will undergo careful review to avoid potential bias or error, and the researchers will identify no limitations in the study.

I. Transparency in Reporting

The study's findings will be reported honestly, with no manipulation or alteration of data. All sources, including academic references, will be correctly cited and

acknowledged to ensure respect for intellectual property rights and uphold ethical research standards. The researchers will utilize AI tools to support various aspects of the research.

J. Use of Patented or Copyrighted Materials

The researchers will not use any patented materials in this study. All copyrighted materials, including references, will be cited correctly in accordance with intellectual property laws. The researchers will ensure they appropriately acknowledged all sources, maintaining academic integrity and avoiding plagiarism.

K. Considerations for Human Subjects

The researchers will inform the respondents about the study's possible risks and benefits and will ensure that they provided an informed consent form. If any significant risks will be associated, such as emotional, physical, or psychological, an appropriate health monitoring program and emotional support services will be available.

L. Responsible Use of AI and Related Technology

The researchers will utilize digital technologies throughout the research process, including finding reliable articles to maintain the accuracy and credibility of the study. AI tools, such as ChatGPT, will be used to assist in summarizing statements and enhancing the quality of the study. The researchers will use platforms like Google Scholar, Elicit, and Typeset.io to find supporting studies, ensuring the reliability and credibility of the research. Also, Grammarly will ensure the research report adheres to proper grammar and clarity standards. The responsible use of technologies will help maintain the integrity of the study.

M. Ethical Approval

The research will obtain ethical approval from the Institutional Review Board of Saint Michael College of Caraga, confirming compliance with all relevant guidelines regarding respondents' protection, data privacy, and intellectual property rights.

RESEARCH INSTRUMENTS

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FOOD SAFETY PRACTICES OF STREET FOOD VENDORS IN BUENAVISTA AND NASIPIT

Part I: Demographic Profile. This part focuses on establishing the demographic profile of participants gathering information on age, sex, and educational attainment.

Direction: Put a check (√) on the box that corresponds to your answer.

NAME (Optional): _____

Age:

- ☐ 18-25 years old
- ☐ 26-34 years old
- ☐ 35 years old and above

Sex:

- ☐ Female
- ☐ Male

Educational Attainment:

- ☐ No Formal Education
- ☐ Elementary Graduate
- ☐ High School Graduate
- ☐ College Graduate

PART II.

Directions: Using the rating scale below

Please put a Check (√) inside the box that indicates your level of agreement for each statement.

(Palihug markahi ug tsek sa lingin nga nahiangay sa imong tubag.)

Scale	Verbal Description	Verbal Interpretation
4	**Strongly Agree (SA)**	**-The practice meets the highest standards.** *(Ang pamaagi nakatagbo sa labing taas nga standard.)*
3	**Agree (A)**	**-The practice is fairly maintained.** (Ang pamaagi medyo maayo ug gisunod.)
2	**Disagree (D)**	**-The practice is poorly maintained.** (Ang pamaagi dili maayo ug kulang sa sunod.)
1	**Strongly Disagree (SD)**	**-The practice fails to meet standards.** (Ang pamaagi wala makatagbo sa standard.)

Part II: This part focuses on the food safety practices of street food vendors in Buenavista and Nasipit, particularly their sanitation practices and personal hygiene.

Sanitation Practices	**SA (4)**	**A (3)**	**D (2)**	**SD (1)**
1. Always clean and sanitize the kitchen tools and equipment and store it properly using shelving unit. (Kanunayon ang paglimpyo ug sanitisa nga mga gamit sa kusina og iplastar sa insaktong butanganan.)				
2.Always clean and sanitize work areas, cutting boards, knives and utensils. (Kanunayon ang paglimpyo ug sanitisa sa lugar nga gitrabahuan, taparan, kutsilyo og uban pang mga gamit.)				
3.Always use the serving utensils with long handle when-stirring large amount of food. (Kanunayon ang paggamit og luwag nga nay taas nga haweranan og magkaray og dinaghan nga niluto nga pagkaon.)				

4.Always use gloves or utensils to handle food after cooking. (Kanunayon ang paggamit og gloves kung mag-alsa o maghandle og pagkaon nga humana na gi luto.)				
5. Always cook the food properly within the required time at the required temperature. (Kanunayon ang pagluto sa pagkaon sa sulod sa sakto og girekomenda nga oras og temperature.)				
6. Minimizing bare hands contact with food. (Minusan 0 likayan ang paghikap sa pagkaon labi na kung walay gloves.)				
7.Checking the availability of hand sanitizer. (Kanunayon ang pagsusi kung aduna ba'y saktong supply sa hand sanitizer.)				
8. Always clean and check sink for washing utensils. (Kanunayon ang pagsusi og paglimpyo sa lababo alang sa paghugas sa mga gamit.)				
9. Avoid smoking in the kitchen while preparing food. (Ginadili ang pagpanigarilyo sa kusina samtang nag andam ug pagkaon.)				
10. Always clean and store the kitchen tools and equipment properly. (Kanunayon ang paglimpyo og insaktong pagplastar sa mga pang kusina.)				

Personal Hygiene	**SA (4)**	**A (3)**	**D (2)**	**SD (1)**
1. Check the self-health condition, body temperature every working day. (Pagsusi sa kahimtang ug kahimsog sa kaugalingon sama sa kainiton sa lawas kung aduna ba'y hilanat ug pagkalibang matagb adlaw nga nag trabaho.)				
2. Checking cleanliness of clothes, hair restraints and prescribed shoes before working. (Pagsusi sa kalimpyo sa sinina, pagpugong sa buhok, pagsuot ug sapatos ayha magsugod sa magtrabaho.)				
3. Washing hands before handling foods. (Pagpang-hugas sa mga kamot sa dili pa muhikap sa mga pagkaon.)				
4. Checking long and colored finger nails which could contaminate food with food-bome illness. (Pagsusi sa tag-as ug gikuloran ba ang mga kuko sa kamot nga maoy mahimong hinungdan nga magdala ug sakit sa pagkaon.)				
5. Wearing of gloves is necessary when handling food especially those who have abrasion or cuts in their hands. (Gikinahanglan ang pagsuot ug gwantes kung muhikap sa mga pagkaon labina kung adunay samad sa ilang kamot.)				
6. Wearing of accessories like rings, watches and bracelets is prohibited when handling food. (Ginadili ang pagsul-ob ug mga aksesorya sama sa mga singsing, relo, ug polseras kung maghimo sa pagkaon.)				
7. Avoid eating in the kitchen while preparing foods. (Ginadili ang pagkaon sa kusina samtang nag andam sa mga pagkaona.)				
8. Wearing of mask while preparing foods is an important practice to reduce the risk of food				

contamination. (Pagsuot ug maskara kung nag-andam ug pagkaon maoy usa ka importante nga buluhaton aron malikay ang pagkaon sa kontaminasyon.)				
9. Wearing of PPE is an important practice to prevent the risks in the workplace and to avoid food contamination. (Ang pagsuot sa saktong sapot pangkusina maoy importanteng buluhaton aron ma malikayan ang disgrasya sa gi-trabahoan ug masugpo ang kontaminasyon sa pagkaon.)				
10. Avoid always talking with co-worker while preparing food. (Ginadili ang pag sige'g estorya sa kauban kung naghikay sa pagkaon.)				

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Membership/Affiliations

ORGANIZATION	POSITIONS HELD
SLICE	**MEMBER**

CHAPTER 3

RESULTS AND DISCUSSION

3.1 Presentation and Analysis of Data

3.2 Proposed Intervention/ Output

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3.2 IS OPTIONAL IF THERE IS INTERVENTION PROGRAM

CHAPTER 4

SUMMARY, CONCLUSION, RECOMMENDATION,

4.1. Summary

4.2. Conclusion

4.3.Recommendation

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APPENDICES

CERTIFICATE OF AUTHENTIC AUTHORSHIP